

Week 3 Task 1: Implementing a Basic Machine Learning Model Using Python

1. Introduction

This project demonstrates an end-to-end machine learning workflow using the Titanic dataset. The objective is to build a baseline classification model capable of predicting passenger survival based on selected features.

2. Dataset

Dataset: Titanic Dataset (publicly available on Kaggle). The dataset contains passenger demographics, ticket details, and survival status. The target variable is **Survived**.

3. Data Preprocessing

The dataset was loaded into a Jupyter Notebook using pandas. Missing values in the Age column were replaced with the median, missing Embarked values were filled with the mode, unnecessary columns such as PassengerId, Name, Ticket, and Cabin were removed, and categorical variables were converted into numerical format using one-hot encoding.

4. Feature Selection and Train-Test Split

Input features were separated from the target variable. The dataset was split into training and testing sets using an 80/20 ratio with `train_test_split` from Scikit-learn.

5. Model Implementation

A Logistic Regression classifier was selected as the baseline model. The model was trained using the training dataset and then used to predict survival outcomes for the testing dataset.

6. Model Evaluation

Performance was evaluated using Accuracy Score, Classification Report (Precision, Recall, and F1-score), and a Confusion Matrix. These metrics provide insight into the model's predictive performance and classification quality.

7. Results

The Logistic Regression model produced strong baseline performance on the Titanic dataset, demonstrating that passenger-related features are useful for predicting survival. The confusion matrix and classification metrics confirmed that the model generalized well to unseen test data.

8. Limitations

The model is a baseline implementation and may not achieve maximum predictive performance. Additional feature engineering and hyperparameter tuning could improve accuracy.

9. Future Improvements

Potential improvements include trying Decision Trees, Random Forests, Gradient Boosting, cross-validation, and advanced feature engineering techniques.

10. Conclusion

This project successfully implemented a complete machine learning pipeline including preprocessing, feature selection, model training, testing, and evaluation. It provides a strong foundation for more advanced machine learning projects.